

## INDIAN HEAD, SK, Canada

WMO: 715150

Lat: 50.550N

Lon: 103.650W

Elev: 1900

StdP: 13.71

Time Zone: -6.00 (W06)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -26.8      | -22.2 | -33.0                         | 0.9 | -26.6 | -28.0 | 1.2 | -22.1 | 32.4                  | 19.6 | 29.8 | 16.6 | 8.7                      | 290  | 0.748 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.0                   | 85.9            | 66.1 | 82.3 | 64.9 | 79.1 | 63.3 | 70.9                | 79.5 | 68.3 | 77.4 | 66.1 | 75.2 | 12.8                 | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.1                            | 110.7 | 75.7 | 65.2 | 99.8 | 73.0 | 62.8 | 91.6 | 70.7 | 36.0          | 79.5 | 33.8 | 77.4 | 31.9 | 75.0 | 86.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 27.9          | 24.9 | 22.1 | DB                         | -34.6 | 92.6               | 4.9 | 3.8                                                | -38.1 | 95.3       | -41.0 | 97.5       | -43.7 | 99.7       | -47.3 | 102.5 | (4) |
| (5)               |      |      | WB                         | -34.8 | 75.7               | 4.9 | 3.4                                                | -38.3 | 78.1       | -41.2 | 80.1       | -43.9 | 82.0       | -47.4 | 84.4  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 36.3   | 5.9   | 8.5   | 20.2  | 37.4  | 50.3  | 59.4  | 63.8  | 62.6  | 53.6  | 39.4  | 23.1  | 9.8   |  |  |
|                                                                        | DBStd       | 23.85  | 15.73 | 14.38 | 14.38 | 10.96 | 8.88  | 6.46  | 5.49  | 6.43  | 8.40  | 9.51  | 12.84 | 14.72 |  |  |
|                                                                        | HDD50       | 6418   | 1367  | 1162  | 924   | 392   | 107   | 6     | 0     | 1     | 56    | 349   | 807   | 1247  |  |  |
|                                                                        | HDD65       | 10607  | 1832  | 1582  | 1389  | 827   | 461   | 188   | 89    | 124   | 351   | 795   | 1257  | 1712  |  |  |
|                                                                        | CDD50       | 1422   | 0     | 0     | 0     | 15    | 116   | 289   | 428   | 392   | 163   | 19    | 0     | 0     |  |  |
|                                                                        | CDD65       | 137    | 0     | 0     | 0     | 0     | 5     | 21    | 52    | 50    | 9     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 2173   | 0     | 0     | 0     | 7     | 142   | 304   | 657   | 800   | 252   | 11    | 0     | 0     |  |  |
|                                                                        | CDH80       | 566    | 0     | 0     | 0     | 0     | 28    | 67    | 145   | 255   | 70    | 1     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 10.7   | 11.5  | 10.8  | 11.2  | 11.3  | 11.8  | 10.5  | 8.8   | 8.8   | 10.3  | 11.0  | 11.0  | 11.0  |  |  |
|                                                                        | PrecAvg     | 17.5   | 0.9   | 0.7   | 1.0   | 1.1   | 2.2   | 2.7   | 2.6   | 2.0   | 1.7   | 1.1   | 0.7   | 0.9   |  |  |
|                                                                        | PrecMax     | 24.3   | 1.9   | 2.3   | 3.3   | 4.0   | 5.1   | 7.5   | 7.1   | 6.7   | 3.8   | 2.9   | 1.8   | 2.0   |  |  |
| (15) Precipitation                                                     | PrecMin     | 9.4    | 0.2   | 0.1   | 0.1   | 0.1   | 0.2   | 0.4   | 0.4   | 0.1   | 0.2   | 0.0   | 0.1   | 0.2   |  |  |
|                                                                        | PrecStd     | 3.4    | 0.4   | 0.5   | 0.7   | 0.9   | 1.3   | 1.6   | 1.7   | 1.6   | 1.1   | 0.9   | 0.4   | 0.4   |  |  |
|                                                                        | DB          | 38.9   | 39.4  | 55.1  | 73.8  | 83.6  | 86.7  | 88.1  | 91.4  | 86.9  | 74.6  | 58.1  | 42.1  |       |  |  |
|                                                                        | MCWB        | 35.5   | 36.0  | 45.9  | 52.3  | 59.2  | 66.1  | 70.7  | 66.4  | 62.7  | 55.5  | 46.0  | 36.1  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 35.8   | 35.4  | 48.2  | 66.7  | 78.1  | 81.0  | 84.0  | 86.5  | 81.3  | 66.3  | 48.8  | 37.1  |       |  |  |
|                                                                        | 2% MCWB     | 33.2   | 33.1  | 40.8  | 49.7  | 55.7  | 62.8  | 69.0  | 65.6  | 61.0  | 51.6  | 41.3  | 34.0  |       |  |  |
|                                                                        | 5% DB       | 32.3   | 32.3  | 42.0  | 61.4  | 73.0  | 77.3  | 80.7  | 82.3  | 75.8  | 59.7  | 43.5  | 33.5  |       |  |  |
|                                                                        | 5% MCWB     | 30.7   | 31.0  | 37.8  | 46.8  | 53.7  | 61.3  | 68.2  | 64.7  | 58.6  | 48.2  | 38.2  | 31.5  |       |  |  |
|                                                                        | 10% DB      | 28.0   | 28.8  | 37.3  | 56.1  | 68.2  | 73.6  | 77.6  | 78.3  | 70.3  | 54.2  | 39.2  | 29.5  |       |  |  |
|                                                                        | 10% MCWB    | 26.8   | 27.6  | 34.8  | 44.5  | 51.6  | 59.7  | 66.2  | 63.1  | 56.1  | 45.6  | 35.8  | 28.0  |       |  |  |
|                                                                        | 0.4% WB     | 36.0   | 36.6  | 46.1  | 54.6  | 62.1  | 71.5  | 76.0  | 72.0  | 66.4  | 57.4  | 47.0  | 36.7  |       |  |  |
|                                                                        | 0.4% MCDB   | 38.6   | 39.2  | 53.4  | 69.0  | 74.6  | 80.3  | 83.2  | 81.5  | 79.3  | 68.7  | 57.2  | 40.7  |       |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 33.4   | 33.3  | 41.7  | 51.1  | 58.2  | 66.5  | 72.5  | 69.0  | 62.9  | 52.7  | 41.9  | 34.1  |       |  |  |
|                                                                        | 2% MCDB     | 35.7   | 35.0  | 47.4  | 64.0  | 71.3  | 75.3  | 80.1  | 80.3  | 76.7  | 64.7  | 47.9  | 36.7  |       |  |  |
|                                                                        | 5% WB       | 30.6   | 31.0  | 37.8  | 48.3  | 55.7  | 63.9  | 69.8  | 66.9  | 60.2  | 49.2  | 38.6  | 31.4  |       |  |  |
|                                                                        | 5% MCDB     | 32.3   | 32.4  | 41.6  | 59.2  | 68.3  | 72.5  | 78.0  | 78.0  | 72.5  | 57.7  | 42.9  | 33.6  |       |  |  |
|                                                                        | 10% WB      | 26.8   | 27.7  | 35.0  | 45.5  | 53.5  | 61.6  | 67.5  | 64.8  | 57.3  | 46.4  | 35.8  | 28.0  |       |  |  |
|                                                                        | 10% MCDB    | 27.8   | 28.8  | 37.2  | 54.6  | 65.5  | 70.0  | 75.4  | 75.0  | 68.4  | 53.7  | 39.1  | 29.6  |       |  |  |
|                                                                        | MDBR        | 17.7   | 17.9  | 18.1  | 22.1  | 24.9  | 21.5  | 23.0  | 25.8  | 24.8  | 20.5  | 17.0  | 16.3  |       |  |  |
|                                                                        | 5% DB       | 18.1   | 16.0  | 20.5  | 31.8  | 32.4  | 28.3  | 27.8  | 32.3  | 33.8  | 30.3  | 23.3  | 18.2  |       |  |  |
| (36) Mean Daily Temperature Range                                      | MCWBR       | 16.6   | 14.1  | 14.5  | 17.3  | 15.0  | 13.9  | 15.5  | 14.7  | 15.5  | 16.9  | 16.1  | 15.7  |       |  |  |
|                                                                        | 5% WB       | 18.4   | 15.6  | 19.4  | 29.1  | 27.1  | 22.7  | 25.0  | 27.7  | 30.0  | 28.4  | 21.9  | 17.9  |       |  |  |
|                                                                        | MCWBR       | 16.9   | 13.9  | 14.1  | 16.8  | 13.9  | 13.2  | 15.9  | 14.9  | 14.9  | 16.4  | 15.4  | 15.6  |       |  |  |
|                                                                        | MCWBR       | 16.9   | 13.9  | 14.1  | 16.8  | 13.9  | 13.2  | 15.9  | 14.9  | 14.9  | 16.4  | 15.4  | 15.6  |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.247  | 0.268 | 0.273 | 0.332 | 0.343 | 0.353 | 0.365 | 0.378 | 0.327 | 0.282 | 0.251 | 0.241 |       |  |  |
|                                                                        | taud        | 2.311  | 2.278 | 2.453 | 2.379 | 2.402 | 2.410 | 2.392 | 2.341 | 2.466 | 2.555 | 2.492 | 2.354 |       |  |  |
|                                                                        | Ebn at Noon | 255    | 278   | 300   | 289   | 290   | 286   | 281   | 271   | 276   | 274   | 258   | 241   |       |  |  |
|                                                                        | Edh at Noon | 20     | 28    | 28    | 35    | 35    | 35    | 36    | 36    | 28    | 21    | 17    | 17    |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 403    | 734   | 1162  | 1546  | 1845  | 1898  | 2052  | 1630  | 1227  | 718   | 391   | 286   |       |  |  |
|                                                                        | RadStd      | 35     | 68    | 89    | 103   | 157   | 141   | 116   | 127   | 104   | 90    | 58    | 25    |       |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (12 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page